

CIS 106 Session Assignments Set 1 Problems.

Develop an IPO for the following problems. Put them into this Word document and save. Upload the document to your GitHub Repository. Finally, paste the link to your repository into the assignment upload link in Blackboard for grading.

Save your files with the convention PS2P1, PS2P2, etc. (see syllabus for details on the naming convention).

1. Prompt the user to enter a quantity (which is a floating point number) and price per unit (float). Then the computer extended price (quantity x price per unit). Display the extended price.

Input	Process	Output
Quantity	Quantity X PPI	Extended price
PPI		

2. Allow the user to enter last name, hours, and pay rate. Compute gross pay as hours x rate. (Note: we are not giving time and a half for overtime hours yet!). Display last name and gross pay.

Input	Process	Output
Last Name of Employee	Accounting for the <i>Last Name of Employee</i> , calculate gross pay for hours worked (<40).	Last Name, Gross Pay.
Hours Accrued		
Pay Rate		

3. The user is to enter the length and width of a rectangle. Compute the area (length x width) and the circumference ($2 \times \text{length} + 2 \times \text{width}$). Display the area and circumference.

Input	Process	Output
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Length of Rectangle	Using the <i>Length of Rectangle</i> and <i>Width of Rectangle</i> , compute the area using length X width.	Display the area of the rectangle.
Width of Rectangle	Using the <i>Length of Rectangle</i> and the <i>Width of Rectangle</i> , compute the circumference using the rectangle circumference formula.	Display the circumference of the rectangle.

4. Enter last name and credits taken. Tuition is \$250 per credit hour. Add a \$100 lab fee. Compute total tuition (credits taken x 250 + lab fee). Display last name and tuition.

Input	Process	Output
Last Name	Calculate total tuition by multiplying <i>Credits Taken</i> (X) by \$250 per credit hour (Y). Add a \$100 charge (Lab Fee) to the result of (X)(Y).	Display Last name and tuition.
Credits Taken		

5. The price of an item and the discount percent are entered into the program. Display the discount amount and the discounted price of the item. Note: Enter the discount percent in decimal form

Input	Process	Output
Price of Item	<i>Price of Item</i> X <i>Discount Decimal Value</i> are multiplied.	Display the price with the discount.

Discount Decimal Value		Display the amount of money the discount removed.
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